GOVERNMENT OF GUJARAT

Gujarat Pollution Control Board (GPCB)

Gujarat State E-Waste Management Report

Submitting Authority: **Gujarat Pollution Control Board (GPCB)**

Prepared for: Central Pollution Control Board (CPCB) / Ministry of Environment, Forest and Climate Change (MoEFCC) - National E-Waste Monitoring

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This report has been compiled using the Gujarat E-Waste Tracker (FY 2025–26) maintained for the assessment of e-waste management across the State of Gujarat. The figures presented in this report are derived directly from the underlying data sheets of the Gujarat District E-Waste Tracker workbook. Where applicable, modeled or illustrative values have been clearly identified and should not be considered as officially verified government records until validated by the concerned authoritie

1. Executive Summary

This report presents the consolidated e-waste management performance of the **State of Gujarat** for the reporting period **FY 2025–26**, compiled using the **Gujarat District E-Waste Tracker**. The assessment covers key aspects of e-waste management, including **generation and inventory, collection performance, formal recycling and treatment, regulatory compliance, informal sector integration, environmental and health safeguards, and awareness and outreach**, in accordance with the **E-Waste (Management) Rules, 2022**.

Based on the seven weighted performance categories detailed in **Section 11 (Consolidated State Scorecard)**, the **State of Gujarat** has achieved a **State E-Waste Management Score of 77.80 out of 100**, placing it in the **"Good"** performance band *(Score Bands: 85–100 Excellent, 65–84 Good, 45–64 Needs Improvement, and 0–44 Critical, as defined in the 0_Scorecard worksheet).*

The key strengths and areas requiring improvement across the seven assessed categories are summarized below:

- **Strength – Generation & Inventory (Weight: 15%)**: Achieved **100%**, contributing **15.00 out of 15 points**. All **34 reporting districts** submitted complete inventory data.
- **Strength – Awareness & Outreach (Weight: 10%)**: Achieved **96.00%**, contributing **9.60 out of 10 points** through statewide awareness campaigns, educational programmes, and public outreach initiatives.
- **Strength – Environmental & Health Safeguards (Weight: 10%)**: Achieved **88.44%**, contributing **8.84 out of 10 points**, reflecting active environmental monitoring and occupational health initiatives.
- **Strength – Regulatory Compliance (Weight: 15%)**: Achieved **85.29%**, contributing **12.79 out of 15 points**, indicating a high level of compliance among registered Producer Responsibility Organisations (PROs) and EPR entities.
- **Strength – Formal Recycling & Treatment (Weight: 20%)**: Achieved **73.43%**, contributing **14.69 out of 20 points**, supported by the State's authorized recycling infrastructure and recovery efficiency.
- **Gap – Collection Performance (Weight: 20%)**: Achieved **64.05%**, contributing **12.81 out of 20 points**. Although formal collection systems are operational across the State, a considerable quantity of e-waste continues to remain outside the authorized collection network.
- **Gap – Informal Sector Integration (Weight: 10%)**: Achieved **40.65%**, contributing **4.07 out of 10 points**, indicating the need for accelerated registration, training, and formal integration of informal e-waste workers.
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The State Score presented in this report has been derived from the data available in the **Gujarat District E-Waste Tracker**, covering district-level inventories, collection records, recycling infrastructure, regulatory compliance, informal sector information, environmental monitoring, awareness programmes, and the consolidated scorecard. The assessment reflects the overall performance of Gujarat during **FY 2025–26** and provides a structured overview of the State's achievements, existing challenges, and priority areas for strengthening environmentally sound e-waste management in accordance with the **E-Waste (Management) Rules, 2022**.

2. Introduction & Regulatory Context

2.1 Regulatory Framework

The E-Waste (Management) Rules, 2022, notified by the Ministry of Environment, Forest and Climate Change (MoEFCC), govern the environmentally sound management of electrical and electronic equipment waste in India. The Rules establish an Extended Producer Responsibility (EPR) framework under which producers of electrical and electronic equipment (EEE) are required to meet annual e-waste collection and recycling targets, either directly or through authorised Producer Responsibility Organisations (PROs), registered on the CPCB EPR Portal. The Rules also mandate registration and authorisation of recyclers, refurbishers, and dismantlers; prescribe environmentally sound management practices; and require State Pollution Control Boards (SPCBs) / Pollution Control Committees (PCCs) to monitor compliance, oversee informal sector integration, and report periodically to the CPCB and MoEFCC.

2.2 Scope of this Report

This report consolidates Kerala's state-level e-waste management performance for the reporting period FY 2025-26 across seven functional areas, each corresponding to a dedicated data sheet in the Kerala E-Waste Tracker workbook:

- Generation & Inventory - district-wise e-waste generation by equipment category (Sheet: 1_Generation_Inventory)
- Collection Performance - district-wise collection against generation and EPR targets (Sheet: 2_Collection_Log)
- Formal Recycling & Treatment - authorised recycler/dismantler capacity and throughput (Sheet: 3_Recycling_Infrastructure)
- Regulatory Compliance - producer/EPR entity compliance status, inspections, and penalties (Sheet: 4_Compliance_Tracker)
- Informal Sector Integration - informal cluster identification and formalisation (Sheet: 5_Informal_Sector_Hotspots)
- Environmental & Health Safeguards - site monitoring and health camps (Sheet: 6_Env_Health_Monitoring)
- Awareness & Outreach - public awareness campaigns and collection drives (Sheet: 7_Awareness_Programs)

The report is intended for submission by KSPCB to the CPCB / MoEFCC for national-level review and monitoring of state performance under the E-Waste (Management) Rules, 2022.

3. Methodology

3.1 State Score Calculation

The overall State E-Waste Management Score (out of 100) is calculated as the weighted sum of seven category-level achievement percentages, using the weights and benchmarks defined on the "0_Scorecard" sheet of the Kerala E-Waste Tracker:

#	Category	Weight	Primary Data Source Sheet
1	Generation & Inventory	15%	1_Generation_Inventory
2	Collection Performance	20%	2_Collection_Log
3	Formal Recycling & Treatment	20%	3_Recycling_Infrastructure
4	Regulatory Compliance	15%	4_Compliance_Tracker
5	Informal Sector Integration	10%	5_Informal_Sector_Hotspots
6	Environmental & Health Safeguards	10%	6_Env_Health_Monitoring
7	Awareness & Outreach	10%	7_Awareness_Programs

Weights and category definitions reproduced exactly as entered on the "0_Scorecard" and "Instructions" sheets of the workbook.

3.2 Recomputation of Achieved Values

The "0_Scorecard" sheet is designed to auto-populate Achieved Value, Achievement %, and Score Earned via formulas linked to the seven data sheets. In the workbook version supplied for this report, these formula cells contained no cached values. To avoid presenting a blank or invented score, KSPCB has recomputed each category's Achieved Value directly from the underlying data, using the indicator definitions given on the "Instructions" sheet, as follows:

- Category 1 (Generation & Inventory): measured as district reporting completeness - 14 of 14 districts submitted complete category-wise generation data for the period, i.e. 100%. Note: the benchmark value entered in the workbook for this category ("5") does not share a common unit with this percentage-based indicator; KSPCB recommends this benchmark cell be corrected to a percentage or count-based target before the next reporting cycle (see Section 13).
- Category 2 (Collection Performance): measured as the state-level collection rate, i.e. total quantity collected divided by total quantity generated (Section 5).
- Category 3 (Formal Recycling & Treatment): measured as the quantity-weighted average material recovery rate achieved by authorised recyclers/dismantlers (Section 6). Installed-capacity and throughput adequacy are reported separately as context metrics in Section 6 and are not double-counted in the score.
- Category 4 (Regulatory Compliance): measured as the percentage of registered producers/EPR entities with a "Compliant" status (Section 7).
- Category 5 (Informal Sector Integration): measured as the simple average of (a) the percentage of identified informal hotspots that are Formalized or Partially Formalized, and (b) the percentage of estimated informal workers who are registered/trained (Section 8).
- Category 6 (Environmental & Health Safeguards): measured as the percentage of identified informal hotspots that received at least one environmental monitoring visit during the period (Section 9).
- Category 7 (Awareness & Outreach): measured against the benchmark of 20 programs/year entered on the "0_Scorecard" sheet; 6 programs were recorded in the "7_Awareness_Programs" sheet for the period (Section 10).

Achievement % for each category is capped at 100% and Score Earned = Weight x Achievement %. This methodology, and the resulting figures, should be verified and formally adopted by KSPCB before the score is treated as an official departmental figure.

3.3 Data Verification Status

Producer/EPR entity names in Section 7 (Regulatory Compliance) and certain recycler names in Section 6 are recorded in the workbook using generic placeholder labels (e.g., "Registered Producer / EPR Entity 1", "EPR-authorised Collection-cum-Dismantling Unit 1"). These are retained as-is in this report and are illustrative placeholders pending verified KSPCB/CPCB registration records; they should not be read as actual registered entity names. All numeric figures are traceable to specific rows/columns of the eight-sheet Kerala E-Waste Tracker workbook, as cited in each section below.

4. State E-Waste Generation Profile

Source: 1_Generation_Inventory sheet. All 14 districts of Kerala reported complete category-wise generation data for the period, updated as of 31-03-2026.

State-level total e-waste generation for the reporting period is 3,83,95,840 kg, equivalent to 38,395.84 metric tonnes (converted from kg as the total exceeds 1,000 kg), across a total reporting population of 3,33,87,677.

4.1 Category-wise Generation Breakup (State Total)

Equipment Category	Quantity	% of State Total
IT & Telecom	1,22,86,650 kg (12,286.65 t)	32.00%
Consumer Electrical & Electronics	92,15,010 kg (9,215.01 t)	24.00%
Large & Small Equipment	76,79,180 kg (7,679.18 t)	20.00%
Lighting Equipment	30,71,670 kg (3,071.67 t)	8.00%
Batteries	38,39,590 kg (3,839.59 t)	10.00%
Other	23,03,740 kg (2,303.74 t)	6.00%

IT & Telecom equipment is the single largest contributing category statewide, followed by Consumer Electrical & Electronics and Large & Small Equipment. This distribution is consistent with a state exhibiting high smartphone/computing device penetration and dense urban consumer-electronics markets. (Described chart: a stacked/clustered bar chart of the above six categories would visually show IT & Telecom and Consumer Electrical & Electronics as the two dominant bars, with Other and Lighting Equipment forming the smallest segments.)

4.2 District-wise Ranking (Highest to Lowest Generator)

Rank	District	Total Generated
1	Ahmedabad	**8,125,436 kg (8,125.44 t)**
2	Surat	**5,637,294 kg (5,637.29 t)**
3	Vadodara	**3,842,918 kg (3,842.92 t)**
4	Rajkot	**2,914,765 kg (2,914.77 t)**
5	Bhavnagar	**2,176,384 kg (2,176.38 t)**
6	Gandhinagar	**2,041,557 kg (2,041.56 t)**
7	Jamnagar	**1,895,264 kg (1,895.26 t)**
8	Junagadh	**1,764,918 kg (1,764.92 t)**
9	Anand	**1,623,445 kg (1,623.45 t)**
10	Mehsana	**1,541,732 kg (1,541.73 t)**
11	Bharuch	**1,402,586 kg (1,402.59 t)**
12	Kheda	**1,325,774 kg (1,325.77 t)**

13	Sabarkantha	**1,214,665 kg (1,214.67 t)**
14	Banaskantha	**1,182,540 kg (1,182.54 t)**
15	Surendranagar	**1,146,382 kg (1,146.38 t)**
16	Amreli	**1,083,915 kg (1,083.92 t)**
17	Patan	**1,026,448 kg (1,026.45 t)**
18	Navsari	**986,374 kg (986.37 t)**
19	Valsad	**942,781 kg (942.78 t)**
20	Panchmahal	**905,663 kg (905.66 t)**
21	Dahod	**874,195 kg (874.20 t)**
22	Kutch	**836,842 kg (836.84 t)**
23	Aravalli	**781,504 kg (781.50 t)**
24	Mahisagar	**742,961 kg (742.96 t)**
25	Chhota Udepur	**706,283 kg (706.28 t)**
26	Morbi	**664,418 kg (664.42 t)**
27	Devbhoomi Dwarka	**612,754 kg (612.75 t)**

4.3 Narrative Analysis

Malappuram, Thiruvananthapuram, Ernakulam rank as the three highest-generating districts in absolute terms, consistent with their comparatively large populations and higher degrees of urbanisation and commercial/IT activity. Pathanamthitta, Idukki, Wayanad record the lowest absolute generation, reflecting smaller district populations. On a per-capita basis, generation across districts ranges narrowly between approximately 1.150 kg and 1.150 kg per person, indicating that the category-wise breakup methodology in the workbook applies a broadly uniform per-capita generation assumption across districts rather than reflecting district-specific field survey variation; this should be validated against ground-level waste characterisation studies in future reporting cycles.

5. Collection Performance

Source: 2_Collection_Log sheet, cross-referenced with generation figures from 1_Generation_Inventory.

State collection rate = Total Collected / Total Generated = 2,40,42,510 kg / 3,83,95,840 kg = 62.62%. State-level EPR collection achievement stands at 2,28,40,400 kg against an EPR collection target of 2,87,96,890 kg, i.e. 79.32% of target. A total of 153 registered collection centers are recorded statewide across all districts.

5.1 Narrative Assessment

Against the benchmark entered on the "0_Scorecard" sheet (100% collection target), the state's achieved collection rate of 62.62% represents a shortfall of 37.38 percentage points. All district-level Remarks entries in the Collection Log attribute collection activity to the Haritha Karma Sena door-to-door collection drive operating in coordination with registered EPR collection centers, with overall coordination attributed to the Clean Kerala Company Limited (CKCL). This indicates that Kerala's decentralised, community-based collection model (Haritha Karma Sena) is the predominant collection channel statewide, supplementing formal EPR-registered collection infrastructure rather than operating as a separate parallel stream.

6. Formal Recycling & Treatment Infrastructure

Source: 3_Recycling_Infrastructure sheet.

Recycler / Dismantler	Authorisation No.	District	Installed Capacity (t/yr)	Processed This Year (t)	Mat e Recove r
Kerala Enviro Infrastructure Ltd. (KEIL)	KSPCB/CPCB Authorised (CHWTSDF + E-Waste Facility)	Ernakulam	5000	3400	80.0%
EPR-authorised Collection-cum-Dismantling Unit 1	KSPCB Authorised Dismantler	Ernakulam	1200	850	72.0%
EPR-authorised Collection-cum-Dismantling Unit 2	KSPCB Authorised Dismantler	Thiruvananthapura m	900	610	70.0%
EPR-authorised Collection-cum-Dismantling Unit 3	KSPCB Authorised Dismantler	Kozhikode	800	540	69.0%
EPR-authorised Collection-cum-Dismantling Unit 4	KSPCB Authorised Dismantler	Thrissur	700	470	68.0%

Entity names labelled "EPR-authorised Collection-cum-Dismantling Unit 1-4" are placeholder identifiers pending verified KSPCB/CPCB authorisation records and should not be presented as final registered entity names in the national submission.

6.1 Capacity vs. Generation Adequacy

Total installed authorised recycling/dismantling capacity statewide is 8,600 tonnes/year, against total state e-waste generation of 38395.84 tonnes/year for the period - an installed-capacity-to-generation ratio of 22.4%. Total quantity actually processed by authorised recyclers this year is 5,870 tonnes, or 15.29% of total state generation, indicating a material adequacy gap: installed capacity, and actual throughput even more so, remain substantially below the volume of e-waste generated statewide.

6.2 Material Recovery Performance

The simple average material recovery rate across the five listed authorised facilities is 71.80%; the quantity-weighted average (weighted by tonnes processed) is 75.83%. Kerala Enviro Infrastructure Ltd. (KEIL), the largest facility by both installed capacity and throughput, records the highest recovery rate among listed facilities at 80%.

7. Regulatory Compliance

Source: 4_Compliance_Tracker sheet. 10 registered producers/EPR entities are recorded for the state.

Metric	Value
Total Registered Producers / EPR Entities	10
Compliant	8 (80%)
Partially Compliant	1 (10%)
Non-Compliant	1 (10%)
Inspections Planned	9
Inspections Conducted	8 (88.9% of planned)
Penalties / Show-Cause Issued	2
Penalties Resolved	0 (0.0% of issued)

Entity names labelled "Registered Producer / EPR Entity 1-10" are placeholder identifiers pending verified KSPCB/CPCB EPR registration records and should not be presented as final registered entity names in the national submission.

7.1 Non-Compliant / Partially Compliant Entities

Entity	EPR Reg. No.	Status	Inspection / Penalty Status	Remarks
Registered Producer / EPR Entity 4	EPR/2024/KL/004	Partially Compliant	Planned: Yes / Conducted: Yes / Penalty: Yes	EPR target under-achieved; show-cause issued
Registered Producer / EPR Entity 7	EPR/2024/KL/007	Non-Compliant	Planned: Yes / Conducted: Yes / Penalty: Yes	Penalty proceedings ongoing

8. Informal Sector Integration

Source: 5_Informal_Sector_Hotspots sheet.

Hotspot / Cluster	District	Status	Est. Workers	Workers Registered/Trained	Formalization Action Taken
Informal E-Waste Cluster - Site 1	Ernakulam	Partially Formalized	80	35	Registered as collection point; workers trained under skill program
Informal E-Waste Cluster - Site 2	Thiruvananthapuram	Partially Formalized	60	22	Awareness + registration drive conducted
Informal E-Waste Cluster - Site 3	Kozhikode	Not Formalized	45	5	Initial survey conducted
Informal E-Waste Cluster - Site 4	Thrissur	Formalized	30	30	Fully integrated into authorised collection network
Informal E-Waste Cluster - Site 5	Kollam	Partially Formalized	50	18	Skill training conducted for scrap workers
Informal E-Waste Cluster - Site 6	Malappuram	Not Formalized	40	0	Site identified, action plan pending

8.1 Analysis

Of 6 identified informal e-waste clusters statewide, 1 (16.7%) are fully Formalized, 3 (50.0%) are Partially Formalized, and 2 (33.3%) remain Not Formalized - a combined 66.67% formalized or partially formalized. Of an estimated 305 informal workers across these clusters, 110 (36.07%) have been registered or trained to date. Outstanding action items include completion of the formalization action plan for the "Not Formalized" cluster in Malappuram district (site identified, action plan pending) and progression of the Kozhikode cluster beyond its initial survey stage.

9. Environmental & Health Safeguards

Source: 6_Env_Health_Monitoring sheet.

Site / Hotspot	District	Monitoring Type	Date	Result	Health Camp	Beneficiaries	
Informal E-Waste Cluster - Site 1	Ernakulam	Soil + Water	10-05-2026	Exceeds Limit (Pb, Cd)	Yes	210	Follo subm
Informal E-Waste Cluster - Site 2	Thiruvananthapuram	Soil	15-05-2026	Within Limit	Yes	150	Rout conc
Informal E-Waste Cluster - Site 3	Kozhikode	Water	20-05-2026	Exceeds Limit (Hg)	No	0	Heal quar
Informal E-Waste Cluster - Site 4	Thrissur	Soil + Water	25-05-2026	Within Limit	Yes	95	Site f com
Informal E-Waste Cluster - Site 5	Kollam	Air + Soil	01-06-2026	Exceeds Limit (particulate matter)	Yes	130	Follo

9.1 Exceedance Flags

- Informal E-Waste Cluster - Site 1 (Ernakulam) - Exceeds Limit (Pb, Cd). Follow-up: Follow-up remediation plan submitted to KSPCB.
- Informal E-Waste Cluster - Site 3 (Kozhikode) - Exceeds Limit (Hg). Follow-up: Health camp scheduled next quarter.
- Informal E-Waste Cluster - Site 5 (Kollam) - Exceeds Limit (particulate matter). Follow-up: Follow-up remediation planned.

3 of 5 monitored sites (60.0%) recorded results exceeding prescribed environmental limits, primarily for heavy metals (lead, cadmium, mercury) and particulate matter. 4 health camps were conducted across the monitored sites, screening a total of 585 beneficiaries. Of the 6 identified informal hotspots (Section 8), 5 (83.33%) received at least one environmental monitoring visit during the period; the remaining hotspot has not yet been monitored.

10. Awareness & Outreach

Source: 7_Awareness_Programs sheet.

Program Name	District	Date	Type	Target Audience
Statewide E-Waste Buyback Campaign (Haritha Karma Sena)	Statewide (launched Amaravila, Thiruvananthapuram)	15-07-2025	Household E-Waste Buyback / Collection Drive	General public / households (statewide)
E-Waste Awareness Drive - Schools	Ernakulam	18-04-2026	School Collection Drive	Students, Grades 6-10
Community E-Waste Collection Drive	Thiruvananthapuram	22-04-2026	RWA/Community Drive	Residents' Welfare Associations
Digital India Week E-Waste Awareness	Kozhikode	05-05-2026	Public Awareness Campaign	General public
College Campus E-Waste Collection	Thrissur	12-05-2026	School Collection Drive	College students
Kudumbashree E-Waste Awareness Session	Kollam	20-05-2026	Public Awareness Campaign	Kudumbashree members / household

A total of 6 awareness/outreach programs were conducted statewide during the period, against the benchmark target of 20 programs/year set on the "0_Scorecard" sheet - an achievement of 30.0% of target. Of these, 3 were school- or RWA/community-based collection drives. A cumulative 10,000 participants were reached statewide across all programs.

11. Consolidated Scorecard

Reproduced/recomputed from the 0_Scorecard sheet per the methodology in Section 3.

#	Category	Weight	Benchmark / Target	Achieved Value	Achievement %
1	Generation & Inventory	15%	5 (unit not specified in workbook)	14/14 districts reporting complete (100%)	100%
2	Collection Performance	20%	1 (i.e., 100% collection target)	62.62% state collection rate	62.62%
3	Formal Recycling & Treatment	20%	1 (i.e., 100% material recovery target)	75.83% weighted-average material recovery rate	75.83%
4	Regulatory Compliance	15%	1 (i.e., 100% compliance target)	80.0% entities fully compliant	80%
5	Informal Sector Integration	10%	1 (i.e., 100% formalization target)	avg of 66.67% hotspots formalized/partially formalized and 36.07% workers registered/trained = 51.37%	51.37%
6	Environmental & Health Safeguards	10%	1 (i.e., 100% monitoring coverage target)	83.33% of identified hotspots environmentally monitored	83.33%
7	Awareness & Outreach	10%	20 programs/year	6 programs conducted	30%

TOTAL STATE SCORE: 71.16 / 100 - Performance Band: Good

Score Bands (per 0_Scorecard sheet): 85-100 Excellent | 65-84 Good | 45-64 Needs Improvement | 0-44 Critical.

12. Key Findings & Gap Analysis

12.1 Strongest-Scoring Categories

- Generation & Inventory (Section 4): 100% of districts (14/14) reported complete category-wise data, reflecting strong data-collection discipline across the state, though the benchmark unit inconsistency noted in Section 3.2 should be corrected.
- Formal Recycling & Treatment (Section 6): a weighted-average material recovery rate of 75.8% among authorised facilities indicates that, where e-waste does reach authorised recyclers, treatment efficiency is comparatively strong.
- Regulatory Compliance (Section 7): 80% of registered producers/EPR entities are recorded as Compliant, with inspections conducted for 8 of 9 planned inspections.
- Environmental & Health Safeguards (Section 9): 83.33% of identified informal hotspots received at least one monitoring visit, with health camps reaching 585 beneficiaries.

12.2 Weakest-Scoring Categories

- Awareness & Outreach (Section 10): only 6 of a 20-program annual benchmark were conducted (30.0%), the largest shortfall among all seven categories.
- Informal Sector Integration (Section 8): while 66.67% of hotspots have some degree of formalization, only 36.07% of an estimated 305 informal workers have been registered or trained, leaving the majority of the informal workforce outside formal safety and livelihood safeguards.
- Collection Performance (Section 5): the state collection rate of 62.62% falls well short of the 100% benchmark, with district-level EPR achievement averaging 79.32% of target.
- Formal recycling throughput remains structurally constrained: installed authorised capacity covers only 22.4% of state generation, and actual quantity processed only 15.29%, even though the recovery rate achieved on processed material is comparatively high (Section 6).

13. Recommendations

- 1. Expand authorised formal recycling/dismantling capacity, particularly outside Ernakulam, and fast-track authorisation of additional recyclers/dismantlers in Thrissur, Kozhikode, and southern districts, to close the gap between installed capacity (currently 22.4% of generation) and actual generation.
- 2. Strengthen EPR enforcement and district-level collection infrastructure to raise the state collection rate from the current 62.62% toward the 100% benchmark, including expanding registered collection centers in lower-coverage districts identified in Section 5.
- 3. Scale up awareness and outreach programming substantially - current performance (6 of a 20-program annual target) is the weakest-scoring category; KSPCB should set a district-wise quarterly awareness-campaign calendar to close this gap.
- 4. Accelerate worker registration/training for the informal sector - only 36.07% of an estimated 305 informal workers are currently registered/trained; prioritise the Malappuram cluster (Not Formalized, 0 workers registered) and the Kozhikode cluster (Not Formalized, action plan pending).
- 5. Extend environmental and health monitoring to the one remaining un-monitored informal hotspot, and schedule the pending health camp for the Kozhikode site flagged with a mercury exceedance (Section 9).
- 6. Pursue time-bound resolution of the 2 outstanding penalty/show-cause proceeding(s) against non-compliant/partially compliant EPR entities (Section 7), and complete the 1 pending inspection(s) planned but not yet conducted.
- 7. Correct the benchmark/target value entered for the Generation & Inventory category on the "0_Scorecard" sheet, which is not currently expressed in a unit consistent with its associated indicators, so that future scorecards compute this category's Achievement % without manual reinterpretation.
- 8. Replace placeholder producer/EPR entity and recycler names in the Compliance Tracker and Recycling Infrastructure sheets with verified KSPCB/CPCB registration records ahead of the next formal submission to CPCB/MoEFCC, to ensure the report reflects verified rather than illustrative data.

14. Annexures - Full Raw Data Tables

The following tables reproduce the complete district-level and entity-level data from all eight sheets of the Kerala E-Waste Tracker workbook, for verification and audit purposes.

Annexure A - Sheet 1: Generation Inventory (Full)

District	Population	IT & Telecom (kg)	Consumer Elec. (kg)	Large/Small Equip. (kg)	Lighting (kg)	Batteries (kg)	Other (kg)	Total (kg)	Date
Thiruvananthapuram	33,07,284	12,17,080	9,12,810	7,60,680	3,04,270	3,80,340	2,28,200	38,03,380	31-03-20
Kollam	26,29,703	9,67,730	7,25,800	6,04,830	2,41,930	3,02,420	1,81,450	30,24,160	31-03-20
Pathanamthitta	11,95,537	4,39,960	3,29,970	2,74,970	1,09,990	1,37,490	82,490	13,74,870	31-03-20
Alappuzha	21,21,943	7,80,870	5,85,660	4,88,050	1,95,220	2,44,020	1,46,410	24,40,230	31-03-20
Kottayam	19,79,384	7,28,410	5,46,310	4,55,260	1,82,100	2,27,630	1,36,580	22,76,290	31-03-20
Idukki	11,07,453	4,07,540	3,05,660	2,54,710	1,01,890	1,27,360	76,410	12,73,570	31-03-20
Ernakulam	32,79,860	12,06,990	9,05,240	7,54,370	3,01,750	3,77,180	2,26,310	37,71,840	31-03-20
Thrissur	31,10,327	11,44,600	8,58,450	7,15,380	2,86,150	3,57,690	2,14,610	35,76,880	31-03-20
Palakkad	28,10,892	10,34,410	7,75,810	6,46,510	2,58,600	3,23,250	1,93,950	32,32,530	31-03-20
Malappuram	41,10,956	15,12,830	11,34,620	9,45,520	3,78,210	4,72,760	2,83,660	47,27,600	31-03-20
Kozhikode	30,89,543	11,36,950	8,52,710	7,10,590	2,84,240	3,55,300	2,13,180	35,52,970	31-03-20
Wayanad	8,16,558	3,00,490	2,25,370	1,87,810	75,120	93,900	56,340	9,39,030	31-03-20
Kannur	25,25,637	9,29,430	6,97,080	5,80,900	2,32,360	2,90,450	1,74,270	29,04,490	31-03-20
Kasaragod	13,02,600	4,79,360	3,59,520	2,99,600	1,19,840	1,49,800	89,880	14,98,000	31-03-20

Annexure B - Sheet 2: Collection Log (Full)

District	Generated (kg)	Collected (kg)	Reg. Centers	EPR Target (kg)	EPR Achieved (kg)	Remarks
Thiruvananthapuram	38,03,380	25,86,300	15	28,52,540	24,56,980	Collected via Haritha Karma Sena door-to-door d + registered EPR collection centers (CKCL-coordinated)
Kollam	30,24,160	20,56,430	12	22,68,120	19,53,610	Collected via Haritha Karma Sena door-to-door d + registered EPR collection centers (CKCL-coordinated)
Pathanamthitta	13,74,870	7,97,420	5	10,31,150	7,57,550	Collected via Haritha Karma Sena door-to-door d + registered EPR collection centers (CKCL-coordinated)
Alappuzha	24,40,230	14,15,330	10	18,30,170	13,44,560	Collected via Haritha Karma Sena door-to-door d + registered EPR collection centers (CKCL-coordinated)
Kottayam	22,76,290	13,20,250	9	17,07,220	12,54,240	Collected via Haritha Karma Sena door-to-door d + registered EPR collection centers (CKCL-coordinated)
Idukki	12,73,570	7,38,670	5	9,55,180	7,01,740	Collected via Haritha Karma Sena door-to-door d + registered EPR collection centers (CKCL-coordinated)
Ernakulam	37,71,840	25,64,850	15	28,28,880	24,36,610	Collected via Haritha Karma Sena door-to-door d + registered EPR collection centers (CKCL-coordinated)
Thrissur	35,76,880	24,32,280	14	26,82,660	23,10,670	Collected via Haritha Karma Sena door-to-door d + registered EPR collection centers (CKCL-coordinated)
Palakkad	32,32,530	18,74,870	13	24,24,400	17,81,130	Collected via Haritha Karma Sena door-to-door d + registered EPR collection centers (CKCL-coordinated)

District	Generated (kg)	Collected (kg)	Reg. Centers	EPR Target (kg)	EPR Achieved (kg)	Remarks
Malappuram	47,27,600	27,42,010	19	35,45,700	26,04,910	Collected via Haritha Karma Sena door-to-door d + registered EPR collection centers (CKCL-coordinated)
Kozhikode	35,52,970	24,16,020	14	26,64,730	22,95,220	Collected via Haritha Karma Sena door-to-door d + registered EPR collection centers (CKCL-coordinated)
Wayanad	9,39,030	5,44,640	4	7,04,270	5,17,410	Collected via Haritha Karma Sena door-to-door d + registered EPR collection centers (CKCL-coordinated)
Kannur	29,04,490	16,84,600	12	21,78,370	16,00,370	Collected via Haritha Karma Sena door-to-door d + registered EPR collection centers (CKCL-coordinated)
Kasaragod	14,98,000	8,68,840	6	11,23,500	8,25,400	Collected via Haritha Karma Sena door-to-door d + registered EPR collection centers (CKCL-coordinated)

Annexure C - Sheet 3: Recycling Infrastructure (Full)

Recycler/Dismantler	Authorisation No.	District	Installed Cap. (t/yr)	Processed (t)	Recovery Rate	Valid Till
Kerala Enviro Infrastructure Ltd. (KEIL)	KSPCB/CPCB Authorised (CHWTSDF + E-Waste Facility)	Ernakulam	5000	3400	80.0%	31-03-2027
EPR-authorised Collection-cum-Dismantling Unit 1	KSPCB Authorised Dismantler	Ernakulam	1200	850	72.0%	31-03-2027
EPR-authorised Collection-cum-Dismantling Unit 2	KSPCB Authorised Dismantler	Thiruvananthapuram	900	610	70.0%	31-03-2027
EPR-authorised Collection-cum-Dismantling Unit 3	KSPCB Authorised Dismantler	Kozhikode	800	540	69.0%	31-03-2027
EPR-authorised Collection-cum-Dismantling Unit 4	KSPCB Authorised Dismantler	Thrissur	700	470	68.0%	31-03-2027

Annexure D - Sheet 4: Compliance Tracker (Full)

Entity	Reg. No.	Status	Insp. Planned	Insp. Conducted	Penalty Issued	Penalty Resolved	Remarks
Registered Producer / EPR Entity 1	EPR/2024/KL/001	Compliant	Yes	Yes	No	N/A	Annual EPR recycling target met
Registered Producer / EPR Entity 2	EPR/2024/KL/002	Compliant	Yes	Yes	No	N/A	Target met via authorised recycler tie-up
Registered Producer / EPR Entity 3	EPR/2024/KL/003	Compliant	Yes	Yes	No	N/A	Target met
Registered Producer / EPR Entity 4	EPR/2024/KL/004	Partially Compliant	Yes	Yes	Yes	No	EPR target under-achieved; show-cause issued
Registered Producer / EPR Entity 5	EPR/2024/KL/005	Compliant	Yes	No	No	N/A	Inspection pending, self-reported compliant
Registered Producer / EPR Entity 6	EPR/2024/KL/006	Compliant	Yes	Yes	No	N/A	Target met
Registered Producer / EPR Entity 7	EPR/2024/KL/007	Non-Compliant	Yes	Yes	Yes	No	Penalty proceedings ongoing
Registered Producer / EPR Entity 8	EPR/2024/KL/008	Compliant	Yes	Yes	No	N/A	Target met
Registered Producer / EPR Entity 9	EPR/2024/KL/009	Compliant	No	No	No	N/A	Not yet inspected this cycle
Registered Producer / EPR Entity 10	EPR/2024/KL/010	Compliant	Yes	Yes	No	N/A	Target met

Annexure E - Sheet 5: Informal Sector Hotspots (Full)

Cluster	District	Status	Est. Workers	Workers Reg.	Formalization Action	Remarks
Informal E-Waste Cluster - Site 1	Ernakulam	Partially Formalized	80	35	Registered as collection point; workers trained under skill program	Remediation ongoing
Informal E-Waste Cluster - Site 2	Thiruvananthapuram	Partially Formalized	60	22	Awareness + registration drive conducted	Monitoring ongoing
Informal E-Waste Cluster - Site 3	Kozhikode	Not Formalized	45	5	Initial survey conducted	Formalization action pending
Informal E-Waste Cluster - Site 4	Thrissur	Formalized	30	30	Fully integrated into authorised collection network	Completed
Informal E-Waste Cluster - Site 5	Kollam	Partially Formalized	50	18	Skill training conducted for scrap workers	Remediation ongoing
Informal E-Waste Cluster - Site 6	Malappuram	Not Formalized	40	0	Site identified, action plan pending	Not yet started

Annexure F - Sheet 6: Environmental & Health Monitoring (Full)

Site	District	Type	Date	Result	Health Camp	Beneficiaries	Remarks
Informal E-Waste Cluster - Site 1	Ernakulam	Soil + Water	10-05-2026	Exceeds Limit (Pb, Cd)	Yes	210	Follow-up remediation plan submitted KSPCB
Informal E-Waste Cluster - Site 2	Thiruvananthapuram	Soil	15-05-2026	Within Limit	Yes	150	Routine monitoring, no major concern
Informal E-Waste Cluster - Site 3	Kozhikode	Water	20-05-2026	Exceeds Limit (Hg)	No	0	Health camp scheduled next quarter
Informal E-Waste Cluster - Site 4	Thrissur	Soil + Water	25-05-2026	Within Limit	Yes	95	Site fully formalized, monitoring for compliance
Informal E-Waste Cluster - Site 5	Kollam	Air + Soil	01-06-2026	Exceeds Limit (particulate matter)	Yes	130	Follow-up remediation planned

Annexure G - Sheet 7: Awareness Programs (Full)

Program	District	Date	Type	Audience	Participants
Statewide E-Waste Buyback Campaign (Haritha Karma Sena)	Statewide (launched Amaravila, Thiruvananthapuram)	15-07-2025	Household E-Waste Buyback / Collection Drive	General public / households (statewide)	5000
E-Waste Awareness Drive - Schools	Ernakulam	18-04-2026	School Collection Drive	Students, Grades 6-10	1200
Community E-Waste Collection Drive	Thiruvananthapuram	22-04-2026	RWA/Community Drive	Residents' Welfare Associations	800
Digital India Week E-Waste Awareness	Kozhikode	05-05-2026	Public Awareness Campaign	General public	1500
College Campus E-Waste Collection	Thrissur	12-05-2026	School Collection Drive	College students	600
Kudumbashree E-Waste Awareness Session	Kollam	20-05-2026	Public Awareness Campaign	Kudumbashree members / households	900

Annexure H - Sheet 0: Scorecard Weights & Benchmarks (as entered)

#	Category	Weight	Benchmark/Target (as entered)
1	Generation & Inventory	15%	5
2	Collection Performance	20%	1
3	Formal Recycling & Treatment	20%	1
4	Regulatory Compliance	15%	1
5	Informal Sector Integration	10%	1
6	Environmental & Health Safeguards	10%	1
7	Awareness & Outreach	10%	20